

Auteur: Niels Hertoghs
 Studierichting: Informatica
 Oefeningenreeks 2D nr. 10.14

$$\lim_{x \rightarrow 2} \frac{x^3 - x^2 - 8x + 12}{3x^3 - 10x^2 + 4x + 8} = \frac{0}{0}$$

$$\begin{array}{c|ccc|c} -3 & 1 & -1 & -8 & 12 \\ & & -3 & 12 & -12 \\ \hline & 1 & -4 & 4 & 0 \end{array}$$

$$\Rightarrow (x^2 - 4x + 4) = (x - 2)^2$$

$$= \lim_{x \rightarrow 2} \frac{(x + 3)(x - 2)^2}{3x^3 - 6x^2 - 4x^2 + 8x - 4x + 8} = \lim_{x \rightarrow 2} \frac{(x + 3)(x - 2)^2}{3x^2(x - 2) - 4x(x - 2) - 4(x - 2)}$$

$$= \lim_{x \rightarrow 2} \frac{(x + 3)(x - 2)}{3x^2 - 4x - 4}$$

$$\begin{array}{c|cc|c} 2 & 3 & -4 & -4 \\ & & 6 & 4 \\ \hline & 3 & 2 & 0 \end{array}$$

$$= \lim_{x \rightarrow 2} \frac{(x + 3)(x - 2)}{(3x + 2)(x - 2)} = \lim_{x \rightarrow 2} \frac{(x + 3)}{(3x + 2)} = \frac{5}{8}$$

References